
GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
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Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)



Revision History

Rev No.	Date	Description of Revision
R01	27/07/2022	Issued for Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.).
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

Restricted

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-ELE-LST-006

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, GAS, PLANT, ROW, PTS

REFERENCES

Document No.	Description

Table of Contents

1. CONTROL ROOM DB SCHEDULE (DB-01) BBOU	5
2. RTU ROOM DB SCHEDULE (DB-02) BBOU	6
3. GUARD & OFFICE BUILDING DB SCHEDULE (DB-03) BBOU	7
4. CONTROL ROOM DB SCHEDULE (DB-01) ELEPA	8
5. RTU ROOM DB SCHEDULE (DB-02) ELEPA	9
6. GUARD & OFFICE BUILDING DB SCHEDULE (DB-03) ELEPA	10
7. CONTROL ROOM DB SCHEDULE (DB-01) AKABEL	11

CONTROL ROOM DB SCHEDULE (DB-01) BBOU														
DB CIRCUIT NO.	DESCRIPTION	CIRCUIT TAG. NO	NO. OF POINTS/ CIRCUITS	LOAD/POINT S (W)	LOAD/CIRCUIT (W)	LOAD FACTOR	CORRECTED PHASE LOAD (W)			VOLTAGE (V)	CURRENT (A)	MCB/RCBO RATING (A)	CABLE SIZE (mm ²)	REMARK
							R	Y	B					
MCB 01	LIGHTING	DB01/L1	2	44	88	1.0	88	-	-	230	0.48	5	1.5	
RCBO 02	SOCKET	DB01/P1	2	300	600	0.6	-	360	-	230	3.26	20	2.5	
RCBO 03	1.0HP AC	DB01/AC1	1	745	745	1.0	-	-	745	230	4.05	20	4.0	
MCB 04	SPARE						-	-	-	230	0.00	5		
MCB 05	SPARE						-	-	-	230	0.00	5		
MCB 06	SPARE						-	-	0	230	0.00	5		
RCBO 07	SPARE						-	-	-	230	0.00	20		
RCBO 08	SPARE						-	-	-	230	0.00	20		
RCBO 09	SPARE						-	-	-	230	0.00	20		
	SINGLE PHASE LOAD (W)						88	360	745					
	THREE PHASE LOAD (W)			1,193										
	THREE PHASE CURRENT AT 400V, 0.8PF (A)			2.15										
	SPECIFICATION			20A, 8 CIRCUITS TPN, MCB/RCBO DISTRIBUTION BOARD										
	SUPPLY CABLE			4C X 6mm ² CUI/XLPE/PVC/SWA/PVC + 4mm ² EARTH CABLE										
NOTE														
1. All power sockets shall be supplied by RCBOs furnished with 30mA RCD														
2. Neutral and earth cables for final circuits shall inherit the size of the live cable supplying each circuit														
3. Type 2 3P + N surge protective device shall be installed on the switchboard. The device shall have a nominal discharge current (In) ≥ 20 kA/mode, maximum discharge current (Imax) ≥ 65 kA, a maximum continous operating voltage (Uc) ≥ 275V and a voltage protection level(Up) ≤ 1.5kv														
REFERENCES:														

RTU ROOM DB SCHEDULE (DB-02) BBOU

DB CIRCUIT NO.	DESCRIPTION	CIRCUIT TAG. NO	NO. OF POINTS/ CIRCUITS	LOAD/POINT S (W)	LOAD/CIRCUIT (W)	LOAD FACTOR	CORRECTED PHASE LOAD (W)			VOLTAGE (V)	CURRENT (A)	MCB/RCBO RATING (A)	CABLE SIZE (mm ²)	REMARK
							R	Y	B					
MCB 01	LIGHTING	DB01/L1	2	44	88	1.0	88	-	-	230	0.48	5	1.5	
RCBO 02	SOCKET	DB01/P1	2	300	600	0.6	-	360	-	230	3.26	20	2.5	
RCBO 03	1.0HP AC	DB01/AC1	1	745	745	1.0	-	-	745	230	4.05	20	4.0	
MCB 04	SPARE						-	-	-	230	0.00	5		
MCB 05	SPARE						-	-	-	230	0.00	5		
MCB 06	SPARE						-	-	0	230	0.00	5		
RCBO 07	SPARE						-	-	-	230	0.00	20		
RCBO 08	SPARE						-	-	-	230	0.00	20		
RCBO 09	SPARE						-	-	-	230	0.00	20		
	SINGLE PHASE LOAD (W)						88	360	745					
	THREE PHASE LOAD (W)			1,193										
	THREE PHASE CURRENT AT 400V, 0.8PF (A)			2.15										
	SPECIFICATION			20A, 8 CIRCUITS TPN, MCB/RCBO DISTRIBUTION BOARD										
	SUPPLY CABLE			4C X 6mm ² CU/XLPE/PVC/SWA/PVC + 4mm ² EARTH CABLE										

NOTE

1. All power sockets shall be supplied by RCBOs furnished with 30mA RCD
2. Neutral and earth cables for final circuits shall inherit the size of the live cable supplying each circuit
3. Type 2 3P + N surge protective device shall be installed on the switchboard. The device shall have a nominal discharge current (I_n) ≥ 20 kA/mode, maximum discharge current (I_{max}) ≥ 65 kA, a maximum continuous operating voltage (U_c) ≥ 275 V and a voltage protection level(U_p) ≤ 1.5 kv

REFERENCES:

GUARD & OFFICE BUILDING DB SCHEDULE (DB-03) BBOU

DB CIRCUIT NO.	DESCRIPTION	CIRCUIT TAG. NO	NO. OF POINTS/ CIRCUITS	LOAD/POINT S (W)	LOAD/CIRCUIT (W)	LOAD FACTOR	CORRECTED PHASE LOAD (W)			VOLTAGE (V)	CURRENT (A)	MCB/RCBO RATING (A)	CABLE SIZE (mm ²)	REMARK
							R	Y	B					
MCB 01	LIGHTING	DB03/L1	2	44	88	1.0		-	88	230	0.48	5	1.5	
MCB 02	LIGHTING	DB03/L2	4	44	176	1.0	176	-	-	230	0.96	5	1.5	
RCBO 03	SOCKET	DB03/P1	2	300	600	0.6	-	360	-	230	3.26	20	2.5	
RCBO 04	SOCKET	DB03/P2	2	300	600	0.6	-		360	230	3.26	20	2.5	
RCBO 05	1.0HP AC	DB03/AC1	1	745	745	1.0	745	-		230	4.05	20	4.0	
RCBO 06	1.0HP AC	DB03/AC1	1	745	745	1.0	-	745		230	4.05	20	4.0	
MCB 07	SPARE						-	-	-	230	0.00	5		
MCB 08	SPARE						-	-	-	230	0.00	5		
MCB 09	SPARE						-	-	0	230	0.00	5		
RCBO 10	SPARE						-	-	-	230	0.00	20		
RCBO 11	SPARE						-	-	-	230	0.00	20		
RCBO 12	SPARE						-	-	-	230	0.00	20		
	SINGLE PHASE LOAD (W)						921	1,105	448					
	THREE PHASE LOAD (W)			2,474										
	THREE PHASE CURRENT AT 400V, 0.8PF (A)			4.46										
	SPECIFICATION													
	20A, 4 WAYS, 12 CIRCUITS TPN, MCB/RCBO DISTRIBUTION BOARD													
	SUPPLY CABLE													
	4C X 10mm ² CU/XLPE/PVC/SWA/PVC + 6mm ² EARTH CABLE													

NOTE

1. All power sockets shall be supplied by RCBOs furnished with 30mA RCD
2. Neutral and earth cables for final circuits shall inherit the size of the live cable supplying each circuit
3. Type 2 3P + N surge protective device shall be installed on the switchboard. The device shall have a nominal discharge current (In) ≥ 20 kA/mode, maximum discharge current (Imax) ≥ 65 kA, a maximum continuous operating voltage (Uc) ≥ 275V and a voltage protection level(U_p) ≤ 1.5kv

REFERENCES:

CONTROL ROOM DB SCHEDULE (DB-01) ELEPA														
DB CIRCUIT NO.	DESCRIPTION	CIRCUIT TAG. NO	NO. OF POINTS/ CIRCUITS	LOAD/POINT S (W)	LOAD/CIRCUIT (W)	LOAD FACTOR	CORRECTED PHASE LOAD (W)			VOLTAGE (V)	CURRENT (A)	MCB/RCBO RATING (A)	CABLE SIZE (mm ²)	REMARK
							R	Y	B					
MCB 01	LIGHTING	DB01/L1	2	44	88	1.0	88	-	-	230	0.48	5	1.5	
RCBO 02	SOCKET	DB01/P1	2	300	600	0.6	-	360	-	230	3.26	20	2.5	
RCBO 03	1.0HP AC	DB01/AC1	1	745	745	1.0	-	-	745	230	4.05	20	4.0	
MCB 04	SPARE						-	-	-	230	0.00	5		
MCB 05	SPARE						-	-	-	230	0.00	5		
MCB 06	SPARE						-	-	0	230	0.00	5		
RCBO 07	SPARE						-	-	-	230	0.00	20		
RCBO 08	SPARE						-	-	-	230	0.00	20		
RCBO 09	SPARE						-	-	-	230	0.00	20		
	SINGLE PHASE LOAD (W)						88	360	745					
	THREE PHASE LOAD (W)			1,193										
	THREE PHASE CURRENT AT 400V, 0.8PF (A)			2.15										
	SPECIFICATION			20A, 8 CIRCUITS TPN, MCB/RCBO DISTRIBUTION BOARD										
	SUPPLY CABLE			4C X 6mm ² CU/XLPE/PVC/SWA/PVC + 4mm ² EARTH CABLE										
NOTE 1. All power sockets shall be supplied by RCBOs furnished with 30mA RCD 2. Neutral and earth cables for final circuits shall inherit the size of the live cable supplying each circuit 3. Type 2 3P + N surge protective device shall be installed on the switchboard. The device shall have a nominal discharge current (In) ≥ 20 kA/mode, maximum discharge current (Imax) ≥ 65 kA, a maximum continuous operating voltage (Uc) ≥ 275V and a voltage protection level(U _p) ≤ 1.5kv														
REFERENCES:														

RTU ROOM DB SCHEDULE (DB-02) ELEPA

DB CIRCUIT NO.	DESCRIPTION	CIRCUIT TAG. NO	NO. OF POINTS/ CIRCUITS	LOAD/POINT S (W)	LOAD/CIRCUIT (W)	LOAD FACTOR	CORRECTED PHASE LOAD (W)			VOLTAGE (V)	CURRENT (A)	MCB/RCBO RATING (A)	CABLE SIZE (mm ²)	REMARK
							R	Y	B					
MCB 01	LIGHTING	DB01/L1	2	44	88	1.0	88	-	-	230	0.48	5	1.5	
RCBO 02	SOCKET	DB01/P1	2	300	600	0.6	-	360	-	230	3.26	20	2.5	
RCBO 03	1.0HP AC	DB01/AC1	1	745	745	1.0	-	-	745	230	4.05	20	4.0	
MCB 04	SPARE						-	-	-	230	0.00	5		
MCB 05	SPARE						-	-	-	230	0.00	5		
MCB 06	SPARE						-	-	0	230	0.00	5		
RCBO 07	SPARE						-	-	-	230	0.00	20		
RCBO 08	SPARE						-	-	-	230	0.00	20		
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REFERENCES:

GUARD & OFFICE BUILDING DB SCHEDULE (DB-03) ELEPA

DB CIRCUIT NO.	DESCRIPTION	CIRCUIT TAG. NO	NO. OF POINTS/ CIRCUITS	LOAD/POINT S (W)	LOAD/CIRCUIT (W)	LOAD FACTOR	CORRECTED PHASE LOAD (W)			VOLTAGE (V)	CURRENT (A)	MCB/RCBO RATING (A)	CABLE SIZE (mm ²)	REMARK
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RCBO 04	SOCKET	DB03/P2	2	300	600	0.6	-		360	230	3.26	20	2.5	
RCBO 05	1.0HP AC	DB03/AC1	1	745	745	1.0	745	-		230	4.05	20	4.0	
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MCB 09	SPARE						-	-	0	230	0.00	5		
RCBO 10	SPARE						-	-	-	230	0.00	20		
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REFERENCES:

CONTROL ROOM DB SCHEDULE (DB-01) AKABEL

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